

ABELARDO ENRIQUE MONSALVE COBIS

EU Citizenship.

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ABOUT ME

Mathematician and Statistician with over 10 years of experience in university teaching, applied research, and statistical and mathematical modeling. I have worked at institutions like BCAM and ITMATI on predictive models, longitudinal data analysis, and interpretable AI. Skilled in R, Python, MATLAB, with strong command of HTML, CSS, and JavaScript. My work bridges academic rigor with practical applications in health, industry, and technology.

KEY SKILLS & COMPETENCIES

- Statistical Computing (R, RStudio, R Markdown, Quarto, Shiny)
- Reproducible Research & Open Science
- Metadata & Data Management (CSV, JSON, XML, relational DBs)
- MS Office Suite (Excel, Word, PowerPoint) – Advanced
- OS: Windows, Linux.
- Research Design, Quantitative Analysis, Multivariate Modelling
- Scientific Writing & Peer-Reviewed Publishing
- Teaching & Academic Supervision (BSc, MSc, PhD)

LANGUAGES

- **Spanish:** Native
- **English:** Upper-intermediate (B1+/B2 – working proficiency, no formal certification)

WORK EXPERIENCE

- **Research Technician**, Basque Center for Applied Mathematics-BCAM, Basque Country, March 2020-June 2024.

Responsibilities

- Data analysis, debugging and visualization in health and clinical studies.
- Statistical analysis and modeling in health and clinical studies.
- Collaboration in interdisciplinary research with medical institutions.

- **Research Technician**, ITMATI - Technological Institute for Industrial Mathematics, Santiago de Compostela, Galicia, January 2019-February 2020.

Responsibilities

- Data cleaning, exploratory analysis, and implementation of statistical methodologies for reliability analysis..
- Developed [lifetime-model-based modules](#) integrated into predictive maintenance systems.

- **Professor**, Universidad Centroccidental Lisandro Alvarado, Lara State, Venezuela, April 2001-January 2019.

Responsibilities

- Teaching class on probability, statistics, stochastic processes, and time series.
 - Supervising undergraduate and doctoral theses.
 - Head of the Department of Operations Research and Statistics (2014–2019).
 - Member of the Doctoral Committee in Mathematics.
- **Research Intern**, PDVSA-INTEVEP - Venezuelan Institute of Oil Technology (1999-2001), Venezuela.

Responsibilities

- Developed multifractal models for the classification of lithological and geophysical data.
 - Contributed to industrial statistical modeling for petroleum research.

ACADEMIC BACKGROUND

- **Bootcamp of Data Science. (500 hrs).** The Bridge | Digital Talent Accelerator 2025.
- **Specialization in Deep Learning: Informed Clinical Decision Making**, Coursera-University of Glasgow, April 2023.
- **PhD in Statistics and Operations Research**, University of Santiago de Compostela, September 2011.
- **Advanced Studies Diploma**, University of Santiago de Compostela, September 2008.
- **Master's in Random Models**, Venezuelan Institute for Scientific Research, February 2003.
- **Bachelor's in Mathematics**, University of Los Andes, September 1998.

SPECIALIZED COURSES

- **Applied Machine Learning Using Python**, 150 hours, State Digital Transformation Plan, July 2022.
- **Deep Learning in Electronic Health Records and Healthcare Applications**, University of Glasgow, December 2022-March 2023.
- **Data Mining for Clinical Databases**, University of Glasgow, November 2022.
- **Applied Artificial Intelligence in Business**, G12 Business Services Group, January-April 2022.
- **Big Data Architecture**, 165 hours, Hedima, Madrid, October-December 2022.
- **Massive Data and Data Science Analyst**, INDICE - Catalonia Continuous Training Consortium, 310 hours, May-August 2022.

PROJECT PARTICIPATION

- **BASQUE MODELLING WORKING GROUP'S:** "Mathematical Modeling Applied to Health Project (2020-2024)"
 - Data analysis and development of mortality prediction models in the COVID-19 pandemic
 - Analysis and development of prediction models for the study of Chronic Obstructive Pulmonary Disease (COPD).
- **Predictive Cognitive Maintenance Decision Support System (PreCoM)**

- Objective: Development and testing of a decision-making system for predictive maintenance, capable of identifying and locating faults, assessing their severity, predicting their evolution, evaluating their life span, and issuing alerts for preventive maintenance actions.
 - Website: PreCoM Project
- **PDVSA-ENT: Calculating payment for liquid fuel transportation services and mechanical maintenance services (2016-2017).**
 - Develop a model for calculating compensation for ENT mechanical maintenance personnel, based on equity and social justice principles, recognizing and encouraging individual and collective contributions to service quality and efficiency.
- **CIEPE-Foundation (State Research Center for Experimental Agroindustrial Production) Interlaboratory Results Comparison Program (2014-2015)**
 - Design and develop a web-based control system for Fundación CIEPE to automate the management, processing, and statistical evaluation of the Interlaboratory Results Comparison Program (PCRI), in compliance with ISO-5725 (Covenin 2972) standards.
- **SEMI-PARAMETRIC, FUNCTIONAL, AND SPATIO-TEMPORAL STATISTICAL METHODOLOGIES AND APPLICATIONS (2009-2013) MAESFE**
 - Participation: Collaborated in developing a financial modeling objective for interbank interest rate series in Spain and Europe during the pre- and banking crisis periods.
 - Referencia: MTM2008-03010. Ministerio de Ciencia e Innovación, España. Facultad de Matemáticas, Universidad de Santiago de Compostela 2009-2013
 - IP-Principal: Wenceslao González Manteiga (IP) <http://eio.usc.es/pub/MAESFE>.

SUPERVISED WORKS

- **Co-Director of Doctoral Thesis:** Pedro Harmath, “Analysis and Estimation of Unidimensional Poverty Measures under Empirical Bootstrap Process Theory”, Universidad Centroccidental "Lisandro Alvarado", Barquisimeto, 2018.
- **Director of Bachelor’s Final Project:** Dellys Matos, “ROC Methodology in Evaluating Cardiovascular Risk Factors as Markers of Epicardial Adipose Tissue. Application to Population”, Universidad Centroccidental "Lisandro Alvarado", Barquisimeto, 2015.
- **Director of Bachelor’s Final Project:** Rafael González, “Point Processes, Theory, and Practical Applications”, Universidad Centroccidental "Lisandro Alvarado", Barquisimeto, 2014.
- **Director of Bachelor’s Final Project:** Karla Carrero, “A Vector Autoregressive Model for Modeling Inflation in Venezuela”, Universidad Centroccidental "Lisandro Alvarado", Barquisimeto, 2013.

PUBLICATIONS

- Cristobal Esteban, Irantzu Barrio; Amaia Aramburu; Abelardo Monsalve-Cobis; Leyre Chasco; Javier Moraza; Josu Najera-Zuloaga; Susana Aizpiri; Myriam Aburto; Inmaculada Arostegui; Jose Maria

Quintana (2024), *Functional Capacity and Physical Activity. An Instrument for Classification of COPD Patients*. Respiratory Medicine. (In review)

- Vaamonde-Rivas, Manuel, Monsalve-Cobis, Abelardo, Pardo-Fernández, Juan Carlos, & de Uña-Álvarez, Jacobo. (2021). *Software module based on lifetime analysis for a predictive maintenance decision-support system*. <http://doi.org/10.5281/zenodo.4557168>
- Harmath, P., Ramoni, J., Monsalve, A. and Fajardo, J. (2021). *About the Bootstrap Weak Convergence for the Foster-Greer-Thorbecke Poverty Index*. Investigación Operacional. 42(2),174-194. <https://rev-inv-ope.pantheonsorbonne.fr/sites/default/files/inline-files/42221-04.pdf>
- Harmath, P., Ramoni, J. and Monsalve, A. (2020). A Glivenko-Cantelli Bootstrap Theorem for the Foster-Greer-Thorbecke Poverty Index. Revista Colombiana de Matemáticas. 54(2), 159-177. Scopus-Scimago (Q4). Available at: <https://revistas.unal.edu.co/index.php/recolma/article/view/93845>
- Harmath, P., Ramoni, J. y Monsalve, A. (2019). *Un Resultado Glivenko-Cantelli Bootstrap para la Medida de Pobreza de Foster-Greer-Thorbecke*. Artículo in extenso publicado en el Congreso de Matemática Aplicada, Computacional e Industrial - MACI 2019 - S12-Métodos Probabilísticos y Estadísticos. VII(2019), 245-248. Asociación Argentina de Matemática Aplicada, Computacional e Industrial. Disp.en: <https://amcaonline.org.ar/maci/index.php/maci2019/maci/paper/viewFile/5127/366>
- González-Manteiga, W., Zubelli, J.,Monsalve-Cobis, Abelardo E., Febrero-Bande, M. (2017) *"Goodness of Fit Test for Stochastic Volatility Models"*,. *From Statistics to Mathematical Finance. Festschrift in Honour of Winfried Stute*. Springer ISBN 978-3-319-50986-0. pp. 89-104.
- A.E. Monsalve-Cobis, W. González-Manteiga, W. Stute (2017), *"The Statistical Impact of Inflation On Interest Rates"*,. Communications in Statistics-Theory and Methods. Volume 46, Issue 14, ISSN: 0361-0926 .<http://dx.doi.org/10.1080/03610926.2015.1130842>
- Harmath, P., Monsalve, A. y Ramoni, J. (2017). *Medición, Método Axiomático y Medidas de Pobreza: Una Revisión*. *Revista de Matemática de la Universidad del Atlántico*. IV(1), 1-15. Disp.en: <http://investigaciones.uniatlantico.edu.co/revistas/index.php/MATUA/article/view/1756>
- González De Gouveia R, Monsalve-Cobis A, (2017), *"Análisis del Riesgo de Incendios Forestales: Un enfoque basado en Procesos Puntuales"*, Publicaciones en Ciencias y Tecnología. Vol. 11 Nro 1. Pp. 7-16 ISSN: 1856-8890
- González De Gouveia R, Monsalve-Cobis A, Vieira Villareal K,(2015),*"Funciones de Luminosidad de Cuasares, Un enfoque Basado en Procesos Puntuales"*, Publicaciones en Ciencias y Tecnología, Volumen 9, Número 2. pp.105–122. ISSN:1856-8890
- Abelardo Monsalve y Pedro Harmath, (2015). *"Introducción al Análisis de Series de Tiempo con Aplicaciones a la Econometría y Finanzas"*. Ediciones IVIC, Instituto Venezolano de Investigaciones Científicas (IVIC). ISBN 978-980-261-163-8, Caracas, Venezuela. https://www.researchgate.net/publication/295106544_Introduccion_al_Analisis_de_Series_de_Tiempo_con_Aplicaciones_a_la_Econometria_y_Finanzas
- Monsalve-Cobis, A., González-Manteiga, W., and Febrero-Bande, M. (2011), *"Goodness of Fit Test for Interest Rates Models: An Approach based on Empirical Processes"*, Computational Statistics & Data Analysis, 55. Pags 3073–3092. Elsevier Science Bv ISSN: 0167-9473
- David Márquez and Juan R. Jiménez- PDVSA, Abelardo Monsalve (IVIC) (2001)

Facies Recognition Using Multifractal Analysis, Paper presented at the 2001 SEG Annual Meeting, San Antonio, Texas, Paper Number: SEG-2001-0580. <https://doi.org/10.1190/1.1816684>. Society of Exploration Geophysicists:
<https://onepetro.org/SEGAM/proceedings-abstract/SEG01/All-SEG01/SEG-2001-0580/89923>

TECHNICAL TOOLS & LANGUAGES

- R, (RStudio, RMarkdown),
- Python (basic-intermediate) (Jupyter Notebooks)
- Git/GitHub
- LaTeX
- MS Excel (advanced), Word, PowerPoint
- SQL, CSV, JSON
- Windows environments

ADDITIONAL INFORMATION

- EU Citizenship: Yes (Spanish)
- Available to start: September-2025.
- Willing to relocate to Brussels and work onsite.